School of Computer Science Engineering and Information Systems
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AGILE DEVELOPMENT PROCESS AND DEVOPS LAB
ASSESSMENT - 4

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Step 1: Install Required Software
Install the following software: 1. Java Development Kit (JDK) 2. Apache Maven 3. Jenkins 4. Git
Verify installation using:

```
java -version
mvn -version
git --version
```

```
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % java -version
java version "21.0.4" 2024-07-16 LTS
Java(TM) SE Runtime Environment (build 21.0.4+8-LTS-274)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.4+8-LTS-274, mixed mode, sharing)
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % mvn -version
Apache Maven 3.9.9 (8e8579a9e76f7d015ee5ec7bfcdc97d260186937)
Maven home: /opt/maven
Java version: 21.0.4, vendor: Oracle Corporation, runtime: /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home
Default locale: en_IN, platform encoding: UTF-8
OS name: "mac os x", version: "13.4", arch: "aarch64", family: "mac"
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % git --version
git version 2.39.2 (Apple Git-143)
```

Step 2: Create Maven Project Folder
Create a new project folder manually. Example folder name:

my_maven_project

Step 3: Project Structure

```
my_maven_project
│
├── pom.xml
├── src
    └── main
        └── java
            └── com
                └── example
                    └── App.java
```

Step 4: Create pom.xml

Create a file named pom.xml in the root folder

```
pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
                             http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>my_maven_project</artifactId>
    <version>1.0-SNAPSHOT</version>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.8.1</version>
                <configuration>
                    <source>1.8</source>
                    <target>1.8</target>
                </configuration>
            </plugin>
        </plugins>
    </build>

</project>
```

Step 5: Create Java File

Create the file App.java in the following location: src/main/java/com/example/App.java

```
arunimasanjeev@Arunimas-MacBook-Air Desktop % cd my_maven_project
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % mkdir -p src/main/java/com/example
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % nano src/main/java/com/example/App.java
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % ls src/main/java/com/example
App.java
```

```
package com.example;

public class App {
    public static void main(String[] args) {
        System.out.println("Hello Maven Jenkins Project");
    }
}

Save modified buffer (ANSWERING "No" WILL DESTROY CHANGES) ?
                    Y Yes
^C Cancel           N No
```

Step 6: Build Maven Project Locally

Open terminal inside the project folder and run:

mvn clean package

Step 7: Build Output

If build is successful you will see:

BUILD SUCCESS

target/

my_maven_project-1.0-SNAPSHOT.jar

```
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:my_maven_project >-------------------
[INFO] Building my_maven_project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ my_maven_project ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ my_maven_project ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/arunimasanjeev/Desktop/my_maven_project/src/main/resources
[INFO]
[INFO] --- compiler:3.8.1:compile (default-compile) @ my_maven_project ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /Users/arunimasanjeev/Desktop/my_maven_project/target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ my_maven_project ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/arunimasanjeev/Desktop/my_maven_project/src/test/resources
[INFO]
[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ my_maven_project ---
[[INFO] No sources to compile
[INFO]
[[INFO] --- surefire:3.2.5:test (default-test) @ my_maven_project ---
[INFO] No tests to run.
[[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ my_maven_project ---
[[INFO] Building jar: /Users/arunimasanjeev/Desktop/my_maven_project/target/my_maven_project-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[[INFO] BUILD SUCCESS
[[INFO] ------------------------------------------------------------------------
[[INFO] Total time:  0.765 s
[[INFO] Finished at: 2026-03-27T20:15:03+05:30
[[INFO] ------------------------------------------------------------------------
```

Step 8: Push Project to GitHub

Initialize Git repository and push the project:

git init

git add .

git commit -m "Maven Project"

git branch -M main

git remote add origin https://github.com/Ramlingams/my_maven_project.git

git push -u origin main

```
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % git init
Initialized empty Git repository in /Users/arunimasanjeev/Desktop/my_maven_project/.git/
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % git add .
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % git commit -m "Maven Project"
[main (root-commit) 4dead71] Maven Project
 7 files changed, 38 insertions(+)
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 target/classes/com/example/App.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/my_maven_project-1.0-SNAPSHOT.jar
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % git branch -M main
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % git remote add origin https://github.com/ArunimaSanjeev/my_maven_project.git
arunimasanjeev@Arunimas-MacBook-Air my_maven_project % git push -u origin ma
in
Enumerating objects: 23, done.
Counting objects: 100% (23/23), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (23/23), 3.23 KiB | 3.23 MiB/s, done.
Total 23 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/ArunimaSanjeev/my_maven_project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Step 9: Open Jenkins

Open Jenkins in your browser:

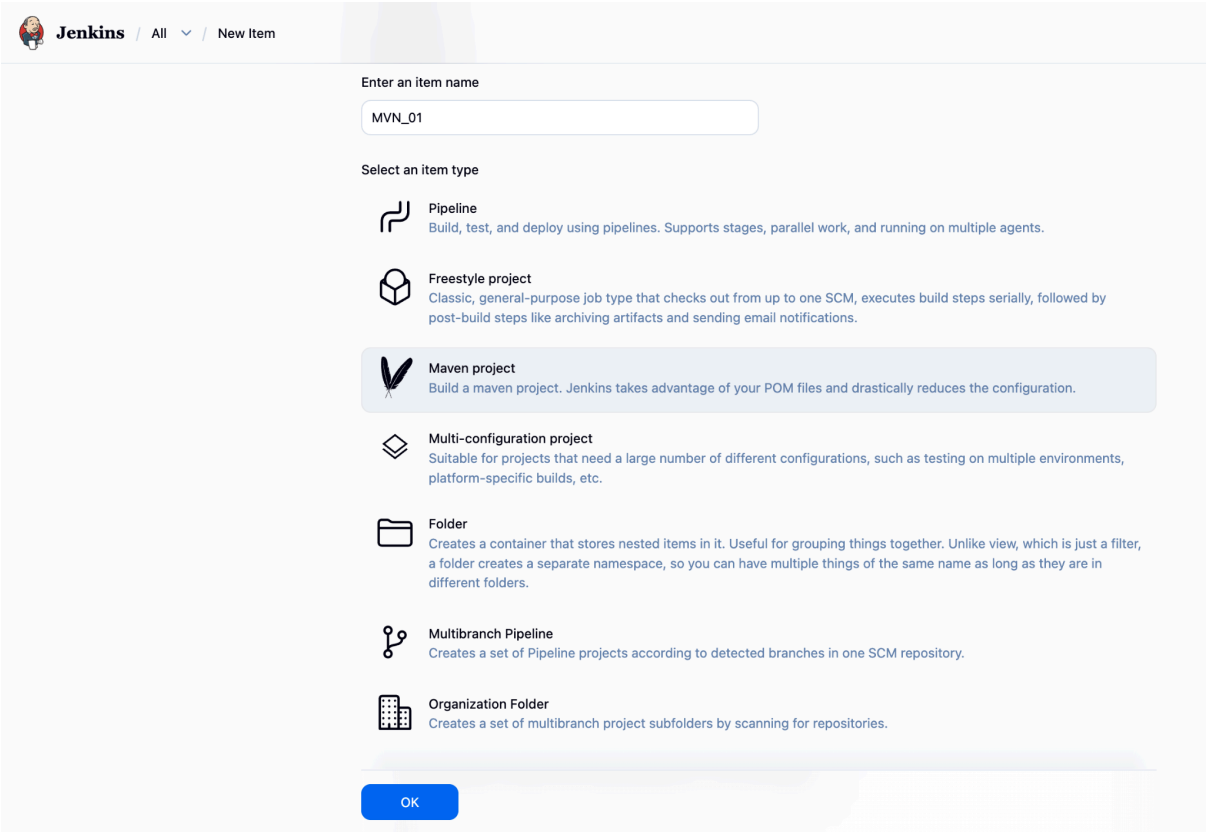
http://localhost:8080

Step 10: Configure Maven in Jenkins

Go to: Manage Jenkins → Global Tool Configuration Add Maven installation: Name: Maven3 Install automatically: ✓

Step 11: Create Maven Project Job

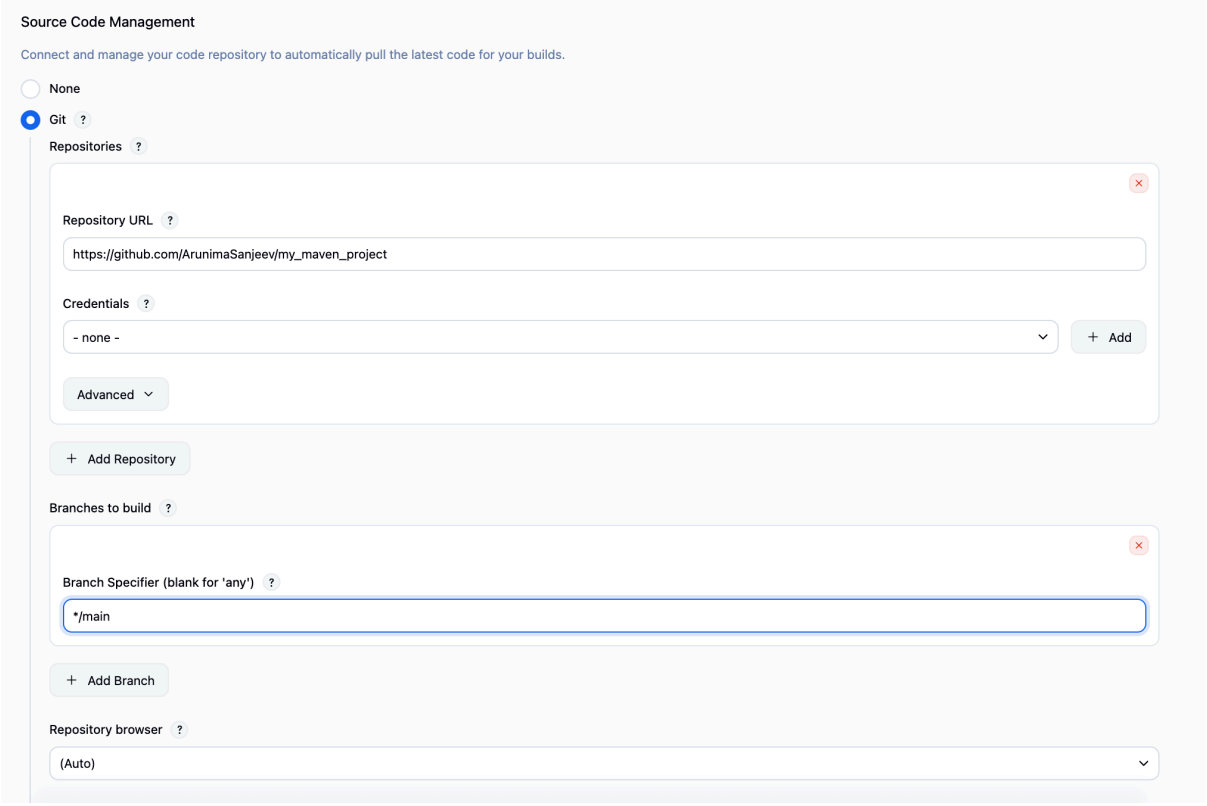
In Jenkins Dashboard: Click New Item -> Enter Name: MVN_01 -> Select: Maven Project -> Click OK



Step 12: Configure GitHub Repository

In Source Code Management select Git and provide:

Repository URL:
https://github.com/Ramlingams/my_maven_project.git
Branch:
*/main



Step 13: Configure Maven Build

In Build section provide:

Root POM : pom.xml
Goals and options : clean package

Step 14: Run Jenkins Build

Click Build Now and open Console Output. Expected result:

BUILD SUCCESS

```
[INFO] Building jar: /Users/arunimasanjeev/.jenkins/workspace/MVN_01/target/my_maven_project-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  16.866 s
[INFO] Finished at: 2026-03-27T20:32:19+05:30
[INFO] ------------------------------------------------------------------------
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving /Users/arunimasanjeev/.jenkins/workspace/MVN_01/pom.xml to com.example/my_maven_project/1.0-
SNAPSHOT/my_maven_project-1.0-SNAPSHOT.pom
[JENKINS] Archiving /Users/arunimasanjeev/.jenkins/workspace/MVN_01/target/my_maven_project-1.0-SNAPSHOT.jar to
com.example/my_maven_project/1.0-SNAPSHOT/my_maven_project-1.0-SNAPSHOT.jar
channel stopped
Finished: SUCCESS
```

Step 15: Final Output

The compiled JAR file will be available in Jenkins workspace:

workspace/MVN_01/target/my_maven_project-1.0-SNAPSHOT.jar